

Task 3 Report

Static Timing Analysis (STA) & Timing Closure of a 4-Bit Carry Lookahead Adder Using Verilog HDL

InternSpark – VLSI Internship

Objective

The objective of this task is to perform Static Timing Analysis (STA) on a synthesized 4-Bit Carry Lookahead Adder (CLA) and analyze the timing characteristics of the digital design using Xilinx Vivado.

Software Used

- * Xilinx Vivado 2025.2
- * Verilog HDL
- * XSim Simulator

Project Description

A Carry Lookahead Adder (CLA) is a high-speed adder architecture that reduces carry propagation delay by generating carry signals in advance. Compared to a Ripple Carry Adder, the CLA provides faster arithmetic operations, making it widely used in high-performance digital systems.

Design Methodology

1. Created the RTL design of a 4-Bit Carry Lookahead Adder.
2. Developed a Verilog testbench for functional verification.
3. Performed behavioral simulation.

4. Synthesized the RTL design using Vivado.
5. Generated the Timing Summary Report.
6. Analyzed timing paths, setup time, hold time, and slack values.

Static Timing Analysis

Static Timing Analysis evaluates whether the synthesized design satisfies timing requirements without applying dynamic input vectors.

The timing report provides information such as:

- * Worst Negative Slack (WNS)
- * Total Negative Slack (TNS)
- * Worst Hold Slack (WHS)
- * Total Hold Slack (THS)

Positive slack indicates that the design satisfies timing requirements, while negative slack indicates timing violations that require optimization.

Timing Closure

Timing closure is the process of ensuring that all timing paths satisfy the specified timing constraints.

Common timing optimization techniques include:

- * Logic optimization
- * Reducing combinational delay
- * Improving placement and routing
- * Register balancing
- * Pipelining for high-speed designs

Results

The 4-Bit Carry Lookahead Adder was successfully synthesized and analyzed using Vivado. Timing reports were generated, allowing verification of setup and hold timing requirements and evaluation of the critical timing path.

Conclusion

The project successfully demonstrated the use of Static Timing Analysis in the digital VLSI design flow. The Carry Lookahead Adder was verified through synthesis and timing analysis, providing practical understanding of timing verification and timing closure techniques used in modern FPGA and ASIC design.